OneSavie Bastet Technical Report
Full-Solution Methodology, Reproducibility, Timeline, and Evidence

Everything Is CTF

July 2026

Abstract

The solution for the OneSavie Bastet Kaggle competition evolved from prior-frequency and code-reading baselines into a public-audit recovery system, a recovered LLM-agent cleanup stage that moved the line from 312.09 to 409.98 public score, then into label-correction and report-recovery pipelines. The final team artifact combines a strong v59 baseline with 40 selected public-report recovery rows, producing the team's documented best score of 519.81 public / 215.21 private. This report emphasizes methodology, reproducibility, rule consistency, and evidence links to the source artifacts.

1 Competition Task and Scope

The task is structured vulnerability identification for smart-contract repositories. The submission row schema is:

`Property, repo_path, severity, tag, subtag, description`.

Submissions are repository-level finding records rather than binary labels. For example, `OneSavieBastet/README.md:3-L5` states that the framework predicts "`(severity, tag, subtag, description)`" for every vulnerability, and `contract-audit-ai/README.md:47-L57` restates the same five semantic fields.

Public references:

- Kaggle competition page: onesavie-bastet.
- Competition landing site: Bastet Kaggle website, which describes the challenge as an LLM smart-contract vulnerability identification competition.
- Official organizer dataset/tooling repository: OneSavieLabs/Bastet, whose README describes Bastet as a dataset of DeFi smart-contract vulnerabilities plus an AI-driven detection process.

1.1 Team and Repositories

The solution was developed under the shared Kaggle account "Everything Is CTF". The kgr experiment file states that the account included `yixuliu`, `hash548`, `matseoizau`, and `wliilamsam`, and that the kgr ledger covers only `yixuliu`'s submissions `OneSavieBastet/final_pack/experiments.md:8-L14`. The methodology is therefore presented as a combined team solution while preserving source-specific attribution where the artifacts explicitly provide it.

Table 1: Solution repositories and their roles

Repository	Role in the combined solution	Evidence
`web3-track`	Early baseline and LLM/Poe work: heuristic baselines, KNN retrieval, official taxonomy alignment, Poe v1/v2, leave-one-repo-out evaluation, and manual audit handoffs.	`web3-track/README.md:11-L20`
`OneSavieBastet`	Main C4/Sherlock lookup framework, dataset_0831 leverage, kgr score ladder, label classifier, description experiments, git-metadata mapping, and post-competition final pack.	`OneSavieBastet/README.md:31-L46`
`contract-audit-ai`	Final report-recovery refinement, deterministic submission hygiene, optional LoRA retagging, confidence scoring, and judge-facing reproduction package.	`contract-audit-ai/README.md:5-L14`

2 Scoring Model and Consequences

The team implemented and reasoned against the same scoring shape across repos. Per repository, predictions are greedily matched to ground-truth rows, and each matched pair receives:

$$\text{PairScore} = \text{TagScore} + \text{SubtagScore} + \text{SeverityScore} + \text{DescriptionScore} - \text{Penalty}.$$

For each structured field:

$$\text{FieldScore} = \max\left(0, \frac{TP - 0.5 \cdot FP}{N}\right).$$

Description similarity counts only above the BGE cosine threshold of 0.7, and over-reporting subtracts a per-repo penalty. This is documented in `contract-audit-ai/README.md:94-L156` and in the agent context scoring notes `web3-track/onesavie-bastet/BASTET_CONTEXT.md:32-L100`.

Important design consequences.

- A submission must remain exactly 400 rows, with `empty` padding if necessary `contract-audit-ai/README.md:185-L202`.
- Over-reporting is expensive; weak row spam hurts all candidate matches in that repo `web3-track/onesavie-bastet/BASTET_CONTEXT.md:73-L86`.
- Once coverage was saturated, same-row label improvements were the cleanest upside: row count, repo count, severity, and descriptions could remain fixed while tag/subtag changed. The v13 gain isolated this effect `OneSavieBastet/daily_training_record.md:459-L466`.
- Description edits could move public score by crossing the BGE threshold, but final private results showed this was not necessarily a generalizing signal `OneSavieBastet/README.md:91-L107`.

3 Data Provenance and Rule Consistency

The solution uses organizer-published competition data and public audit reports, not private leakage. The main README explicitly states that inputs are "officially published by the competition organizer" and include audit reports, source repositories, and ground-truth annotations OneSavieBastet/README.md:10-L16. The final-pack report repeats the same provenance claim OneSavieBastet/final_pack/REPORT.md:41-L49.

The decisive dataset observation was that the 12-character IDs were folder names in the organizer source release. The pipeline then mapped those folders to public Code4rena or Sherlock audits:

> "identify the audit, then transcribe its real findings" OneSavieBastet/final_pack/REPORT.md:89-L93

This is supported by the final-pack workflow description, which lists the data archives, report fetching, contest identification, and 400-row CSV generation OneSavieBastet/final_pack/REPORT.md:151-L190. The contract-audit-ai final method uses the same rule-consistent evidence source: it reads `.git/config` origins, fetches public C4/Sherlock reports, parses High/Medium findings, and inserts only novel rows relative to the baseline contract-audit-ai/docs/report_recovery_pipeline.md:27-L41.

3.1 Artifact Inventory and Count Reconciliation

The artifact counts below were validated directly from the CSV/JSON files used by the solution, rather than copied from prose summaries. This matters because one legacy final-pack table uses a stale train-row count, while the submitted CSVs and the report-recovery CV diagnostic agree on 497 labeled train rows.

Table 2: Competition artifact inventory

Artifact	Rows	Repos	Notes
`train.csv` in all three repos	497	54	365 Medium / 132 High; top tags DoS, Input Validation, Accounting Error.
`test.csv` in all three repos	53	53	Repository IDs only; labels hidden during competition.
v5 pre-agent CSV	400	53	388 non-empty rows; identical to the 312.09 public-score artifact.
post-agent cleanup CSV	400	52	400 non-empty rows; identical to the 409.98 public-score artifact.
v59 baseline CSV	400	52	400 non-empty rows; median description length 280.
final v60 CSV	400	50	400 non-empty rows; median description length 286.5.
v60 selected JSON	40	19	Audit trail for recovered report rows.

The final selected JSON stores `final_score`, `baseline_sim`, `rank`, repository, severity, label fields, label method, and report title contract-audit-ai/submissions/report_recovery_v60_novel40_balanced_selected.json:1-L34. It does not store the full generated descriptions; those are in the final CSV.

4 Method Overview

Figure 1 shows the combined pipeline. The early work starts from train statistics and code heuristics, discovers the public-audit mapping, uses a recovered LLM-agent cleanup stage to reallocate the 400-row budget, then fixes labels and finally replaces weak rows with novel public report findings.

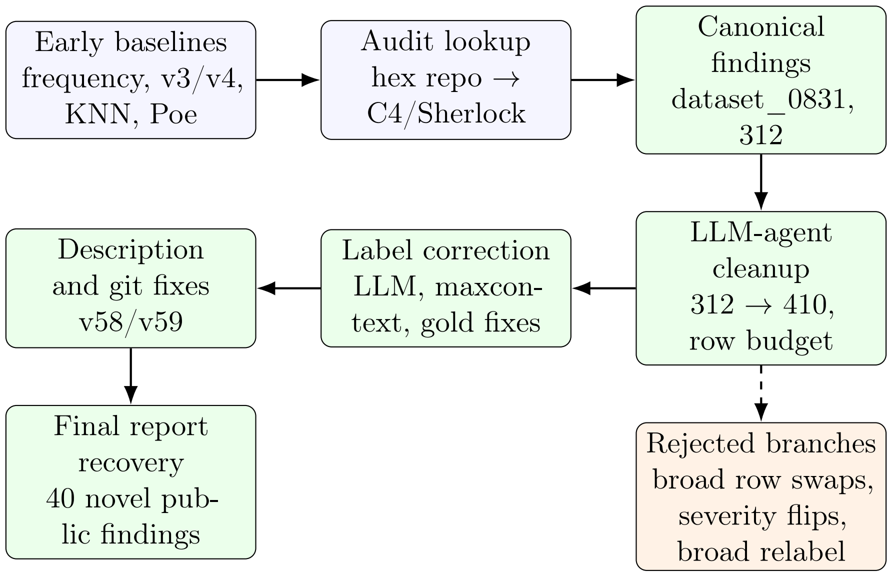

Figure 1: Combined solution flow across the three repositories.

4.1 Stage 0: Early Baselines and LLM/Poe Pipelines

The `web3-track` repository records the first full pipeline family. It evolved from:

- heuristic baselines `build_baseline.py` through v4;
- no-API KNN retrieval over repo text;
- Poe API plus rules, few-shot examples, official taxonomy, and false-positive gating;
- leave-one-repo-out scoring and manual audit handoff tools.

This layered design is summarized in `web3-track/README.md:11-L20`. Poe v2 added a more aggressive coverage budget, median train-derived repo cap, auxiliary text scanning, allocation boost, and train-description overlays `web3-track/README.md:33-L40`.

This stage was important even though it did not become the final high-scoring system. It taught three constraints that remained true later: keep label names exact, avoid over-reporting small repos, and treat holdout validation with caution. The web3-track scoring notes explicitly prefer "fewer, high-quality matches" and `empty` padding over weak spam `web3-track/README.md:41-L48`.

4.2 Stage 1: C4/Sherlock Lookup and dataset_0831

The major jump came when the team mapped repository IDs to public audit contests. The OneSavieBastet final pack describes the lookup pipeline:

1. download organizer-provided `train.zip` and `test.zip`;

2. enumerate Code4rena and Sherlock repositories;
3. fetch public report files;
4. identify each hex folder's audit;
5. parse H/M findings;
6. train a TF-IDF label mapper from train rows;
7. generate the 400-row submission.

The step table is quoted and linked in `OneSavieBastet/final_pack/REPORT.md:181-L190`. The `07_generate_v5_with_dataset0831.py` script adds direct rows from the organizer's public `dataset_0831.csv`: its docstring states that rows covered by that dataset are emitted directly, while the rest fall back to C4/Sherlock lookup and expanded TF-IDF labels `OneSavieBastet/src/pipeline/07_generate_v5_with_dataset0831.py:1-L11`.

4.3 Stage 2: Recovered LLM-Agent Cleanup and Budget Reallocation

The `contract-audit-ai` repository preserves the missing bridge between the 312-point dataset_0831 foundation and the later 442-point coverage skeleton. The pre-agent artifact `submissions/legacy/submission_c4_v5_before_agent.csv` is byte-identical to `OneSavieBastet/data_history/submission_c4_v5_312.09.csv`, and the post-agent artifact `submissions/legacy/submission_v5_agent_cleaned.csv` is byte-identical to `OneSavieBastet/data_history/submission_409.98.csv`. This establishes the score movement for the recovered stage as 312.09 $\rightarrow$ 409.98 public.

The original direct agent service ran on ClawCloud; the provider later became unavailable, so exact prompts and execution logs are not recoverable `contract-audit-ai/docs/judge_method_package.md:56-L65`. The preserved before/after CSVs and comparison tool still show the behavioral shape:

- before: 400 rows, 388 non-empty, 12 `empty`; after: 400 rows, all non-empty;
- 253 rows matched across the before/after comparison: 210 exact, 16 description rewrites, 9 label-only rewrites, and 18 label+description rewrites;
- 135 before rows were likely dropped and 147 after rows were likely added or refill rows;
- row budget moved toward high-value repos, especially `099243e83259` (+31), `0315ba9d8121` (+13), and `fa1836e0615e` (+5), while reducing broad weaker caps.

The repository documents the same conclusion: the lost stage was not a row-aligned retagging pass; it performed row selection, false-positive removal, repo-budget reallocation, empty-row refill, and selective cleanup `contract-audit-ai/docs/judge_method_package.md:80-L91`. The reproducible successor is `scripts/submission_hygiene.py`, which normalizes labels, flags CJK/noisy/duplicate descriptions, checks tag-subtag compatibility, and writes review prompts `contract-audit-ai/docs/submission_hygiene_recovery.md:29-L38`.

4.4 Stage 3: Same-Row Label Classifier

Once the 400 rows were largely real findings, tag/subtag labels became the bottleneck. The key submission was v13:

> "pure tag/subtag correction on the existing 400 rows" OneSavieBastet/daily_training_record.md:463-L466

It used a 3-pass ensemble over guessed rows and overrode labels only where at least two passes agreed OneSavieBastet/daily_training_record.md:457-L461. Later v16 introduced the "maxcontext" prompt: canonical audit text plus extracted code, first identify the code-level defect, then label it OneSavieBastet/final_pack/REPORT.md:197-L216.

This stage is why the kgr line moved from 442.88 to 474.21 public and 145.85 to 191.31 private. The final-pack experiments file calls v13 the largest private lever OneSavieBastet/final_pack/experiments.md:45-L53.

4.5 Stage 4: Gold Fixes, Consensus Gates, and Description Tuning

Several controlled same-row edits improved or clarified the boundary:

- v33 applied three hand-verified dataset_0831 gold corrections and improved private score OneSavieBastet/final_pack/experiments.md:64-L71.
- v46 used five independent Opus passes plus source-code agreement to keep only unanimous, high-confidence label flips OneSavieBastet/final_pack/experiments.md:73-L81.
- v55 dropped six weak rows and added six deduped canonical findings; it became kgr's best private score even though public remained tied OneSavieBastet/final_pack/experiments.md:82-L93.
- v59 rewrote 289 descriptions into concise Cause/Impact form, which became kgr's best public but not the best private OneSavieBastet/final_pack/experiments.md:101-L133.

4.6 Stage 5: Final Public Report Recovery

The `contract-audit-ai` repository packages the final team refinement. It starts from the v59 baseline:

`submissions/baselines/submission_c4_v59_gitfix_descrewrite.csv`

and generates:

`submissions/submission_report_recovery_v60_novel40_balanced.csv`.

The judge-facing package states exactly that final artifact and its 40 selected rows contract-audit-ai/docs/judge_method_package.md:9-L27.

The report-recovery selector is intentionally conservative:

```
final_score = report_rank + label_method_bonus + novelty_bonus
novelty_bonus = max(0, novelty_threshold - baseline_similarity) * 0.35
```

with a 40-row budget, minimum rank 1.18, novelty threshold 0.35, and per-repo caps contract-audit-ai/docs/report_recovery_pipeline.md:43-L62. A train diagnostic matched 482 of 497 train gold rows to public report rows contract-audit-ai/docs/report_recovery_pipeline.md:81-L88, supporting the claim that public audit reports are a strong proxy for hidden target rows.

The final report-recovery artifact has the following properties:

- v59 baseline: 400 non-empty rows, 52 repos, median description length 280 chars.
- final v60 report recovery: 400 non-empty rows, 50 repos, median description length 287 chars.
- semantic diff: 40 rows added and 40 rows dropped relative to v59.
- selected recovery rows: 40 rows; 31 High and 9 Medium; average baseline similarity 0.190, maximum baseline similarity 0.347, average report rank 1.528.
- selected label methods: 28 rule fallback, 6 title-nearest, 4 high-confidence rule, 2 description-nearest.
- repository-count deltas: `03196f805abb` moved from 0 to 4 rows; three one-row baseline repos were fully removed; `d562802cb53a` moved from 5 to 4 rows.

The final method description explains why this can improve score: it recovers missing report findings, rejects already-covered findings, replaces instead of adding rows, and uses report wording for semantic similarity `contract-audit-ai/docs/report_recovery_pipeline.md:114-L125`. The submissions README states the same strategy: the final submission selected public report findings with low semantic overlap against v59 and used them to replace weaker baseline rows `contract-audit-ai/submissions/README.md:21-L26`.

5 Score Timeline

5.1 Repository commit timeline

Table 3: Condensed solution Git history relevant to the writeup

Date	Repo	Commit	Meaning for the solution
2026-06-02	OneSavieBastet	275eb0a / 251931c	Code4rena ground-truth lookup pipeline was introduced.
2026-06-02	OneSavieBastet	295c332 / 23631cd	Sherlock support and extraction fixes were added.
2026-06-04	OneSavieBastet	0e65326 / b246cb2	`dataset_0831` direct lookup and compressed/English variants.
2026-06-11	OneSavieBastet	3d75bac / ffa8b13	Teammate 442 baseline augmentation and measure-then-swap forensics.
2026-06-13	OneSavieBastet	519c787	Era-2 labeling-function classifier moved the line from 442 to 474.
2026-06-24	OneSavieBastet	0c2d536	Opus-teacher relabel and gold corrections documented.
2026-06-27/28	OneSavieBastet	0f8de3a / 2a2ed17	Git metadata mapping, final-day candidates, and v59 best-public description rewrite.
2026-07-01	OneSavieBastet	59b02cf	Retrospective docs reframed around officially provided data and private reveal.
2026-07-01	contract-audit-ai	c271fca	Initial reproducible report-recovery pipeline.
2026-07-02	contract-audit-ai	433a2e5 / e9f54dc	Judge method package and LoRA impact clarification.

Date	Repo	Commit	Meaning for the solution
2026-07-03	web3-track	450185f	English README documenting early Bastet baselines and setup.

5.2 Early public-score history

Table 4: Early score history before the C4/Sherlock system

Date	Submission	Public	Source / read
2026-04-06	`submission.csv`	88.59	prior-based baseline
2026-04-06	`submission_v2.csv`	81.18	over-predicted prior baseline
2026-04-08	`submission_v3.csv`	122.64	best prior-based config
2026-04-11	`first_achieve_v9`	125.88	Poe API + code reading
2026-06-04	`submission_c4_v5.csv`	312.09	dataset_0831 + cached reports
2026-06-04–11	`submission_409.98.csv`	409.98	recovered ClawCloud LLM-agent cleanup
2026-06-11	`submission_c4_v8.csv`	442.88	coverage-maxed teammate base
2026-06-13	`submission_c4_v13_retag.csv`	464.75	LLM tag/subtag classifier
2026-06-14	`submission_c4_v16_maxcontext.csv`	474.21	code-level maxcontext classifier

Source: `OneSavieBastet/daily_training_record.md:5-L32`. The 409.98 row is verified by byte identity between `contract-audit-ai/submissions/legacy/submission_v5_agent_cleaned.csv` and `OneSavieBastet/data_history/submission_409.98.csv`.

5.3 kgr score ladder

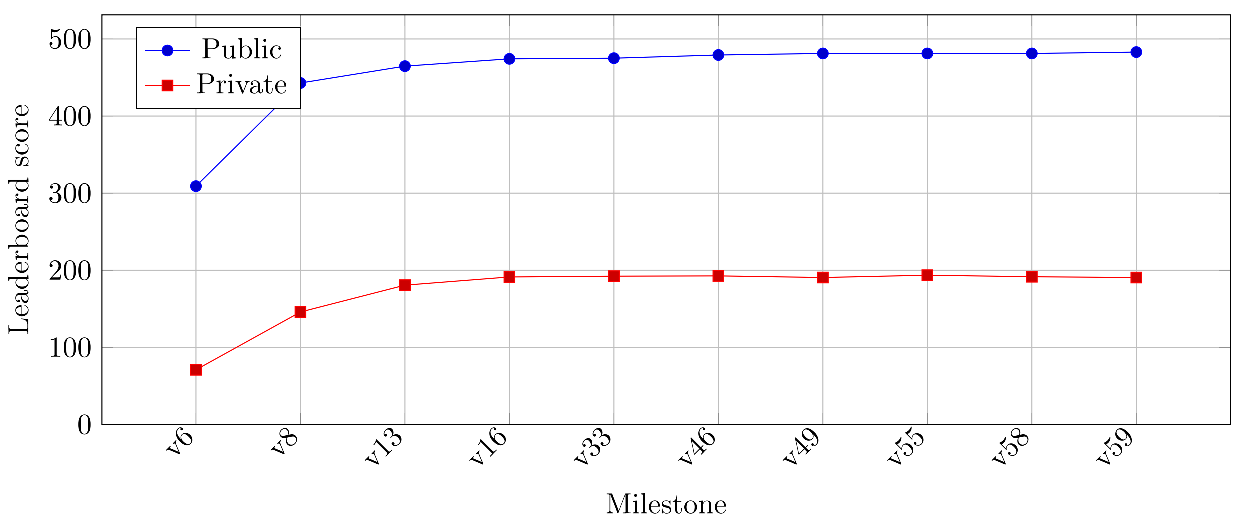

Figure 2: kgr milestone score ladder from the final README and experiments file.

Table 5: kgr milestone attribution

Date	File / milestone	Public	Private	Lever
06-04	v6	309.09	70.89	early C4 lookup
06-11	v8	442.88	145.85	coverage maxed
06-13	v13	464.75	180.67	LLM labeling-function classifier
06-14	v16	474.21	191.31	maxcontext prompt
06-24	v33	475.07	192.31	verified gold tag fixes
06-28	v46	479.21	192.69	5-pass + source consensus
06-28	v55	481.21	193.59	safe coverage swing; kgr best private
06-29	v59	482.97	190.57	concise descriptions; kgr best public

Source: `OneSavieBastet/final_pack/experiments.md:25-L39`.

5.4 Final team artifact

The kgr file explicitly excludes the final top team entry and identifies `report_recovery_v60_novel40_balanced` as a teammate/team submission with score 519.81 public / 215.21 private `OneSavieBastet/final_pack/experiments.md:8-L14`. The main README repeats that the shared Kaggle account's top score was 519.81 / 215.21 and was a teammate entry `OneSavieBastet/README.md:26-L29`. The final artifact is preserved in `contract-audit-ai` and documented in the judge method package `contract-audit-ai/docs/judge_method_package.md:9-L27`.

6 Implementation Map

Table 6: Key code artifacts and functions

Repository	File	Functionality
`web3-track`	`web3-track/onesavie-bastet/build_baseline_v4.py:52-L253`	Pattern bank, Solidity file iteration, entry-point counting, signal detection, per-repo allocation, and baseline generation.
`web3-track`	`web3-track/onesavie-bastet/build_baseline_poe_v2.py:93-L180`	Median train-derived repo cap, allocation boost, auxiliary text scan, train-description overlay, taxonomy normalization, and FP gate.
`contract-audit-ai`	`contract-audit-ai/scripts/compare_submission_versions.py:1-L220`	Compares reordered before/after submissions to reconstruct retained, rewritten, dropped, and added semantic rows for the recovered agent stage.
`OneSavieBastet`	`OneSavieBastet/src/pipeline/07_generate_v5_with_dataset0831.py:1-L126`	Direct dataset_0831 rows plus C4/Sherlock fallback with TF-IDF label transfer.

Repository	File	Functionality
OneSavieBastet	OneSavieBastet/src/validate_submission.py:1-L55	Pre-upload format validator: columns, 400 rows, sequential `Property`, padding counts.
OneSavieBastet	OneSavieBastet/finetune/extract_finding_code.py:1-L41	Extracts matching code snippets from finding descriptions for code-grounded relabel/rewrite passes.
contract-audit-ai	contract-audit-ai/scripts/submission_hygiene.py:86-L523	Deterministic schema, label, duplicate, CJK, style, and review-queue hygiene.
contract-audit-ai	contract-audit-ai/scripts/report_recovery_pipeline.py:160-L1812	Public report fetching/parsing, rule and nearest-neighbor labeling, novelty scoring, train CV, and final 400-row writing.
contract-audit-ai	contract-audit-ai/scripts/retag_submission.py:43-L348	Optional LoRA-assisted label correction with conservative accept/reject rules.
contract-audit-ai	contract-audit-ai/scripts/confidence_evaluator.py:101-L646	Confidence scoring from report support, source-code identifiers, nearest train rows, duplicates, and high repo counts.

7 Reproducibility

7.1 Foundation reproduction

The main OneSavieBastet README gives the foundation reproduction command sequence:

```
pip install -r requirements.txt
python src/pipeline/download_data.py
cd data && unzip -q train.zip && unzip -q test.zip && cd ..
python src/pipeline/01_list_c4_repos.py
python src/pipeline/02_fetch_c4_reports.py
python src/pipeline/03_list_sherlock_repos.py
python src/pipeline/04_fetch_sherlock_reports.py
python src/pipeline/05_identify_contests.py
python src/pipeline/07_generate_v5_with_dataset0831.py
python src/validate_submission.py --submission outputs/submission_c4_v5.csv
```

Source: OneSavieBastet/README.md:149-L173.

7.2 Recovered agent-cleanup audit reproduction

The 312.09 → 409.98 bridge is reproduced as a behavior analysis from preserved CSVs:

```
cd contract-audit-ai
PYTHONPATH=src python scripts/compare_submission_versions.py \
  --before submissions/legacy/submission_c4_v5_before_agent.csv \
  --after submissions/legacy/submission_v5_agent_cleaned.csv \
```

```
  --summary-output outputs/submission_c4_v5_to_submission_summary.json \
  --pairs-output outputs/submission_c4_v5_to_submission_pairs.csv \
  --unmatched-before-output outputs/submission_c4_v5_to_submission_dropped.csv \
  --unmatched-after-output outputs/submission_c4_v5_to_submission_added.csv
```

The command reconstructs row retention, rewrites, likely drops/additions, and repo-count deltas. It does not reproduce the original ClawCloud model execution because those provider-side logs are unavailable `contract-audit-ai/docs/judge_method_package.md:56-L65`.

7.3 Final report-recovery reproduction

The contract-audit-ai README and method package give the final command:

```
uv sync --group dev
python scripts/report_recovery_pipeline.py `
  --target test `
  --fetch-reports `
  --baseline-submission submissions\baselines\submission_c4_v59_gitfix_descrewrite.csv `
  --merge-mode novel-balanced `
  --replace-budget 40 `
  --min-report-rank-for-replace 1.18 `
  --novelty-threshold 0.35 `
  --output-submission outputs\report_recovery_v60_novel40_balanced.csv `
  --preview-output outputs\report_recovery_v60_novel40_balanced_preview.csv `
  --summary-output outputs\report_recovery_v60_novel40_balanced_summary.json
```

Sources: `contract-audit-ai/README.md:20-L45` and `contract-audit-ai/docs/judge_method_package.md:187-L204`.

7.4 Optional LoRA retagging

The LoRA was an auxiliary label-consistency tool, not the final discovery engine. The docs define the task as mapping finding descriptions to exact Bastet tag/subtag label arrays `contract-audit-ai/docs/tag_finetuning.md:3-L10`. Data is converted into chat-format strict JSON examples, with repo-based train/validation split `contract-audit-ai/docs/tag_finetuning.md:11-L51`. Conservative training settings for Gemma 4 12B LoRA are documented in `contract-audit-ai/docs/tag_finetuning.md:70-L84`. The judge package further clarifies that LoRA did not decide row budgets or discover vulnerabilities: it was kept to conservative label cleanup, with 17 correction-mode changes remaining in the final v60 submission `contract-audit-ai/docs/judge_method_package.md:125-L149`.

8 What Did Not Work

Table 7: Exploration branches and negative results

Branch	Evidence	Why it failed or was rejected
Leave-one-repo-out validation as a go/no-go signal	`OneSavieBastet/daily_training_record.md:63-L71`	Holdout repos were biased toward large repos; higher validation score did not imply public score.

Branch	Evidence	Why it failed or was rejected
Description routing from train examples	`OneSavieBastet/daily_training_record.md:171-L173`	Looked good in holdout validation but overfit to holdout auditor language and hurt public score.
v7 LLM source-code augmentation	`OneSavieBastet/daily_training_record.md:339-L345`	A 20-row cap dropped 43 canonical rows; LLM replacements scored near zero.
v9 cached-report additions with proxy gate	`OneSavieBastet/daily_training_record.md:344-L346`	The proxy missed the loss from dropping real true-positive rows.
MISO row swap v12	`OneSavieBastet/daily_training_record.md:423-L436`	Adding uncovered MISO rows required dropping scoring rows; the zero-sum 400-row budget dominated.
Broad report heuristic replacement	`OneSavieBastet/daily_training_record.md:468-L469`	`exp_v20_report_heuristic_gapheavy.csv` collapsed to 48.12 public.
Combined-pool larger few-shot	`OneSavieBastet/daily_training_record.md:545-L560`	More data diluted the targeted prompt; curated maxcontext remained better.
Tree-model tag/subtag operationalizations	`OneSavieBastet/daily_training_record.md:566-L575`	All branch/leaf variants were flat or negative.
Source-code-only full relabel	`OneSavieBastet/README.md:102-L107`	Live-confirmed dead end; source-code-only relabel lost score.
Severity fuzzy flips	`OneSavieBastet/README.md:106-L107`	v61 severity changes lost about 5.3 public points.
Aggressive coverage swing	`OneSavieBastet/final_pack/kgr_kaggle_submissions.csv:7-L7`	v60 bigswing dropped 18 rows and added 18, losing heavily versus v59.
Exact dataset description replacement	`OneSavieBastet/final_pack/kgr_kaggle_submissions.csv:2-L2`	v64 exact dataset text reduced public from 482.97 to 477.27.

9 Evidence Anchors

Table 8: Short source quotes used to anchor the writeup

Claim	Quote	Link
Task definition	"predict the `(severity, tag, subtag, description)`"	`OneSavieBastet/README.md:3-L5`
Organizer-published provenance	"Every input this framework uses is officially published"	`OneSavieBastet/README.md:10-L16`

Claim	Quote	Link
Core audit lookup insight	"identify the audit, then transcribe its real findings"	OneSavieBastet/final_pack/REPORT.md:89-L93
Team top score context	"team's top score (519.81 / 215.21)"	OneSavieBastet/README.md:26-L29
v13 label breakthrough	"pure tag/subtag correction on the existing 400 rows"	OneSavieBastet/daily_training_record.md:463-L466
Final report recovery artifact	"final submitted candidate"	contract-audit-ai/docs/judge_method_package.md:9-L15
Report recovery principle	"only inserts recovered report findings that are likely missing"	contract-audit-ai/docs/report_recovery_pipeline.md:5-L11
LoRA scope	"classification, not vulnerability discovery"	contract-audit-ai/docs/tag_finetuning.md:3-L10
Recovered agent stage boundary	"exact execution logs and prompts could not be recovered"	contract-audit-ai/docs/judge_method_package.md:56-L65
Recovered 312-to-410 mechanism	"not a row-aligned retagging pass"	contract-audit-ai/docs/judge_method_package.md:89-L91
Final selected-row audit trail	"Audit trail of the 40 recovered report findings"	contract-audit-ai/submissions/README.md:8-L14
Early pipeline breadth	"simple frequency baselines into a multi-strategy pipeline"	web3-track/README.md:9-L20
Poe v2 changes	"Higher defaults: target_non_empty=320"	web3-track/README.md:33-L40

A Full kgr Submission Ledger Available in the Repo

The submission ledger is reproduced from OneSavieBastet/final_pack/kgr_kaggle_submissions.csv:1-L41 to document the public/private score timeline. Descriptions are truncated where necessary; exact text is in the linked CSV.

Table 9: kgr authored submissions, public/private scores

Date	File	Public	Private	Note
2026-06-04	submission_c4_v6.csv	309.09	70.89	
2026-06-11	submission_c4_v7.csv	255.61	97.94	
2026-06-11	submission_c4_v8.csv	442.88	145.85	
2026-06-11	submission_c4_v9.csv	410.29	102.64	
2026-06-11	submission_v9_candidate.csv	410.29	102.64	
2026-06-12	submission_probe_P0.csv	420.59	145.85	
2026-06-12	submission_probe_p0_blank_tier_b.csv	420.59	145.85	
2026-06-13	exp_v20_report_heuristic_gapheavy.csv	48.12	137.38	broad report replacement collapse
2026-06-13	submission_c4_v12_miso.csv	424.96	147.22	MISO gap probe, row swap regression
2026-06-13	submission_c4_v13_retag.csv	464.75	180.67	LLM same-row retag breakthrough
2026-06-13	submission_c4_v14_fulltag.csv	468.56	181.30	follow-up fulltag label edit
2026-06-14	submission_c4_v15_canonical.csv	471.18	180.32	aggressive canonical relabel; public up, private down
2026-06-14	submission_c4_v15_subtag.csv	464.62	183.23	conservative subtag-only variant
2026-06-14	submission_c4_v16_maxcontext.csv	474.21	191.31	code-level maxcontext classifier
2026-06-21	submission_c4_v17f_cleandesc.csv	472.16	188.91	clean-description variant
2026-06-21	submission_c4_v17g_cleandesc_short.csv	466.27	184.86	shorter clean descriptions
2026-06-21	submission_c4_v19a_claude_all_best.csv	473.25	183.64	Claude all-row relabel
2026-06-21	submission_c4_v20e_claude_tags_v17e_subtags_cleandesc.csv	456.80	164.06	mixed tags/subtags/descriptions
2026-06-21	submission_c4_v21a_meta_calibrated_all.csv	452.09	174.99	meta-calibrated all rows
2026-06-22	submission_c4_v25d_priority_branch_conf80_baseleaf.csv	474.21	186.59	tree/branch style label variant
2026-06-22	submission_c4_v28a_goldalign_v25d_lowonly.csv	474.21	187.59	gold-alignment low-only variant
2026-06-22	submission_c4_v29j_labeltop49_all6_pair_majority_ge5.csv	473.21	191.31	label-top majority variant
2026-06-22	submission_c4_v30g_labeltop49_gpt55_top49_exclude_v29jloss.csv	472.26	190.58	label-top exclusion variant
2026-06-22	submission_c4_v30h_labeltop49_all6_pair_majority_ge4_exclude_v29jloss.csv	472.24	190.79	label majority threshold variant
2026-06-24	submission_c4_v33_ee25fix_gold.csv	475.07	192.31	three verified dataset_0831 gold tag corrections
2026-06-24	submission_c4_v34_teacher_all.csv	479.28	187.93	full teacher relabel probe
2026-06-25	submission_c4_v40_multitag.csv	478.28	187.93	restored selected second tags

Date	File	Public	Private	Note
2026-06-25	submission_c4_v42_srccode_full.csv	470.88	190.64	full source-code-aware relabel
2026-06-28	submission_c4_v46_5pass_srccode_consensus.csv	479.21	192.69	unanimous five-pass plus source-code agreement
2026-06-28	submission_c4_v47_goldresolve.csv	479.96	188.97	two verified gold corrections
2026-06-28	submission_c4_v49_report_grounded.csv	481.21	190.59	report-grounded corrections
2026-06-28	submission_c4_v50_v49_plus216.csv	481.21	191.59	v49 plus ERC4626/Rounding correction
2026-06-28	submission_c4_v55_swing_safe.csv	481.21	193.59	safe 6-row coverage swing; kgr best private
2026-06-29	submission_c4_v58_gitfix.csv	481.21	191.59	git-metadata wrong-contest fixes
2026-06-29	submission_c4_v59_gitfix_descrewrite.csv	482.97	190.57	concise Cause/Impact rewrites; kgr best public
2026-06-29	submission_c4_v60_bigswing.csv	473.25	182.12	aggressive 18-row swing; failed
2026-06-29	submission_c4_v61_golddesc_sev.csv	477.66	188.50	gold descriptions plus severity flips; failed
2026-06-30	submission_c4_v61b_golddesc.csv	482.66	190.57	compressed gold-row descriptions
2026-06-30	submission_c4_v62b_tight289.csv	481.93	190.43	tightened 289 guessed descriptions
2026-06-30	submission_c4_v64_exactds.csv	477.27	190.57	exact dataset_0831 description replacement

B Artifacts for Verification

Artifact	Repository path
Final CSV	`contract-audit-ai/submissions/submission_report_recovery_v60_novel40_balanced.csv`
Final selected-row audit trail	`contract-audit-ai/submissions/report_recovery_v60_novel40_balanced_selected.json`
Baseline used by final stage	`contract-audit-ai/submissions/baselines/submission_c4_v59_gitfix_descrewrite.csv`
Recovered 312.09 → 409.98 agent-cleanup artifacts	`contract-audit-ai/submissions/legacy/submission_c4_v5_before_agent.csv`; `contract-audit-ai/submissions/legacy/submission_v5_agent_cleaned.csv`
Judge-facing method package	`contract-audit-ai/docs/judge_method_package.md:1-L229`
Foundation report and kgr experiment ledger	`OneSavieBastet/final_pack/REPORT.md:1-L260`; `OneSavieBastet/final_pack/experiments.md:1-L155`